

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 4th Year 2nd Term Examination, 2019.

URP 4273

(Urban Governance)

Full Marks: 210

Time: 3.0 hrs

N.B.: i) Answer any **three** questions from each section in separate script

ii) Figures in the right indicate marks

Section – A

1. (a) What do you mean by Government and Governance? Mention the eight (10) elements of Good Governance according to Dr. Dhiraj Kumar Nath.
(b) Governance has three major components: that of process, contents and (10) deliverables. – Explain.
(c) Define Complex Adaptive System. With the help of a real world context, (15) explain the four characteristics of CAS in a shifting decision making scenario.
2. (a) In light of the events during the Cold War period, make a critical evaluation of (15) the geo-politics centering around the Liberation War of Bangladesh in 1971.
(b) Discuss the major characteristics of the four main types of governance on the (20) basis of Exploration and Exploitation.
3. (a) Critically evaluate the roles of civil society and NGOs to establish good (15) governance in Bangladesh.
(b) Why is Good Governance in the fore-front of global development agenda? (10)
(c) If governance is a multi-stakeholders arrangement, list out the major (10) stakeholders of land for housing in Dhaka city from the governance perspective.
4. (a) Distinguish between the roles and impacts of a Flexible (network based) (10) governance and a State-dominated governance system.
(b) Define the terms ‘Political Correctness’ and ‘Actual Correctness’ with special (15) •relevance to the contemporary socio-political scenario of Bangladesh.
(c) “Whether an institution is playing the role of Government or Governance can (10) be understood by knowing who the head of that institution is.” – Evaluate the statement.

Section – B

5. (a) “Governance: It goes beyond the Government”. Explain it with a practical (15) example?
(b) Suppose, you are working as a Transport Planner in a regional highway (10) development project. Which governance principles you have to consider and maintain for good governance? Explain.
(c) Briefly describe the modes of survey data collection. (10)

6. (a) Suppose Mr. X leads a strike against his authority with the general people. (15)
After a certain period authority replies that “ we should do everything you want” but not in written document. There is no option of sharing the idea to authority. Now draw a ladder showing the participation of citizens and indicate the place where Mr. X and his people stay. Give proper explanation.
- (b) What are the common worldwide governance indicators? Which type of (10)
indicator is necessary for the following situation for assessment?
- To find out massive level of corruption
 - Applying Geo-informatics on foreign affairs of government
 - % access of journalists in govt. working process
 - Level of public participation in rural development
- (c) Provide appropriate examples to describe the various levels of governance. (10)
7. (a) Show the differences between UGI and WGI. (10)
- (b) Calculate and explain the UGI for the following data: (10)
- Effectiveness sub-index = 0.54
 - Equity sub-index = 0.63
 - Participation sub-index = 0.47
 - Accountability sub-index = 0.91
- (c) How does a local government promote inclusive city development? (10)
- (d) A score of -0.45 WGI value and 15th percentile is achieved for control of (05)
corruption in Bangladesh after aggregating indicators. Provide your understanding of the results.
8. (a) What are the major advantages and limitations of Citizen Report Cards? (10)
- (b) “Local Governance Index is organized like a tree” – Explain it with proper (15)
explanation.
- (c) Explain the inherent relationship between the “market & state” in urban (10)
governance as actors?

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 4th Year 2nd Term Examination, 2019

URP 4223

(Urban and Regional Economics)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Explain the Burgess Model, Hoyt Model and Multiple Nuclei Model in brief. 10
Which of these three models, in your opinion, describes the spatial distribution of Khulna city? Why?
- (b) Suppose a person have two income options (per month): to earn \$ 75 US 20
from home based freelancing and working at an office for \$ 140 US (ignore time cost). If one way transportation to office costs \$ 0.25 US/km at 5 km distance, and considering 25 days office days a month, explain his/her decision to accept employment between two options. Also explain, how will his/her decision change in the following conditions:
 - (i) Suppose the office offers transportation cost \$ 30 US per month (ignore time cost);
 - (ii) Suppose it takes 1 hr to reach office and time cost is \$ 0.10 US/hr.
- (c) Why do large cities have higher incomes? 05
2. (a) "City landuse is determined by Bid Rent Theory"- provide argument to 15
refuse or accept the statement.
- (b) What is Demand Cone? Explain how volume of a demand Cone equals to a 13
firm's total revenue.
- (c) A city has three stage of development: innovation, transition and 07
standardization. Why do industries locate in the rural or overseas during standardization stage?
3. (a) Suppose a fast growing city has a high housing demand while it allows only 15
40,000 units per year and the equilibrium price per unit in the inner city area is 10 million BDT. Considering the demand, the city authority decides to expand housing supply by 30,000 additional units per year. Geographically explain the impact on the price change and estimate the monetary value of such permission.
- (b) Following table shows net benefit of four location choices for an industry with 20
three growth scenario. Calculate the equal probability matrix, weighted probability payoff matrix and regret matrix. Considering the regret matrix, which location choice would you suggest for that industry (assume data whatever necessary)?

Payoff Matrix (in millions)			
	Recession	Moderate Growth	Boom
Site - 1	60	40	25
Site - 2	30	45	60
Site - 3	20	50	70
Site - 4	20	67	30

4. Write short note (any five): 5x7=35
- Phase of Urban Sprawl
 - Gentrification
 - Smart Growth
 - Sub – urbanization of office employment.
 - Urban village
 - Euler's Theorem
 - Lorenz curve

Section –B

5. (a) Define Region. Explain the main features of different types of regions with examples. 10
- (b) Briefly explain the concentric model of industrial theory. Use necessary examples and diagrams. 15
- (c) Make a comparison between one-point economy and the spatially extensive economy. 10
6. (a) Narrate the relationship between Keynesian Regional Multiplier and Regional Development. 10
- (b) Explain how to calculate the basic employment using Location Quotients. 07
- (c) How do agglomeration economics contribute to urban economic growth? 08
- (d) Describe the concept of "Product Life Cycle theory" with help of a case study. 10
7. (a) The following table shows different industry's employment of three regions for a nation. Calculate minimum share and basic multiplier separately for each industry for any two regions. 20
- | | Employment | | | |
|------------------------------------|------------|--------|--------|--------|
| Major Industrial Sectors | North | South | West | East |
| Agriculture, forest & fishing | 2526 | 8180 | 4074 | 7847 |
| Mining | 139 | 449 | 1259 | 697 |
| Construction | 34885 | 42000 | 28994 | 55146 |
| Manufacturing | 113441 | 117830 | 179362 | 179691 |
| Transportation | 71652 | 34675 | 57856 | 61430 |
| Wholesale trade | 72719 | 46391 | 78606 | 67643 |
| Retail trade | 138622 | 187673 | 157176 | 153268 |
| Finance, insurance and real estate | 85452 | 66320 | 68927 | 72523 |
| Services | 266781 | 311783 | 365064 | 239308 |
| Non classifiable | 805 | 748 | 759 | 2744 |
- (b) Explain and discuss the limitation of Economic Base Theory. 07
- (c) Write down the importance of Shift Share Analysis on Regional Economy. 08
8. (a) Describe the impact of trade theories on the economy of a nation. 08
- (b) Discuss how production possibilities for products can be calculated under conditions of absolute advantage. Use example if necessary. 15
- (c) What is meant by intensive and extensive farming? 05
- (d) Explain the limitations of New Trade Theory. 07

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Examination, 2019
URP 4217
Coastal and Special Zone Planning

Full Marks: 210

Time: 3.0 hrs

- N.B.: i) Answer any **three** questions from each section in separate script
ii) Figures in the right margin indicate marks
iii) Assume data if necessary

Section – A

Marks

1. (a) 'Different landform formation is a result of costal processes i.e. accretion and erosion'- Discuss this statement. Also depict five types of landforms with necessary diagrams. (15)
- (b) Mention all the basic elements and equations associated with wave theory. (08)
- (c) Discuss various types of wave actions. Also show different wave breakers. (12)
2. (a) What are the common manmade and natural wave action control measures? Provide necessary diagrams. (08)
- (b) Sedimentation process as well as sediment yield depends on several factors. What are those factors? (05)
- (c) Mention some soil erosion management practices. (07)
- (d) Determine annual soil loss from the following information: (15)

Factors	Area A	Area B	Area C
Area (acres)	1856	1670	1587
R factor	255	170	290
Soil type	Clay loam (0.32)	Silt loam (0.69)	Ida silt (0.33)
Topographic slope (%)	12	4	0.8
Topographic length (feet)	600	500	400
Cover management	0.003	0.10	0.00001
Conservation practice	0.60	0.40	No practice

If the slope steepness further increases to 6% for all the areas maintaining the same topographic length, find the changes in soil erosion for area A, B and C.

3. (a) Describe the significance of Tidal River Management (TRM) and polder installation for coastal zone management in Bangladesh. (13)
- (b) What are the targeting processes and targeting groups according to Coastal Development Strategy, 2006? (12)
- (c) Which types of measures should be used to erase wave reflection and diffraction problems? Discuss in brief. (10)
4. (a) Write short notes on the followings (any four): (20)
 - i) Energy Flux model ii) Sediment typology iii) Rill and Gully erosion iv) Spring and reape tide v) Sediment yield.
- (b) Find out (15)
 - (i) Flood Return Period
 - (ii) Draw cross section of the embankment and also find out its area based on the following information:
 - a. Highest Flood Level (HFL) = 16.10m
 - b. Average ground level of river bank = 12.13m
 - c. Free board = 75cm
 - d. Crest width = 3m
 - e. Riverside slope 1:5 and countryside slope 1:3
 - f. Setback distance = 35m
 - g. Probability of occurrence = 0.09.

Section – B

5. (a) Illustrate the concept of coastal zone from multiple perspectives. (08)
(b) Discuss the coastal process, systems and components with necessary example. (15)
(c) Discuss various critical coastal management issues. (12)
6. (a) How does ICZM differ with conventional coastal zone management? Explain with necessary example. (10)
(b) Discuss the characteristics of ICZM for a country having coastal zone. (12)
(c) Discuss various steps of coastal zone planning with necessary example. (13)
7. (a) What is delta? Classify deltas with necessary example. (10)
(b) Discuss five important types of coasts with necessary example. (15)
(c) Draw your concept on:
i) Fringing reef ii) Barrier reef iii) Atolls iv) Patch reef. (10)
8. (a) Describe the importance of ICZM in respect of Bangladesh. (10)
(b) Which types of policy failure may appear in CZM? (10)
(c) Discuss the vulnerability of Bangladesh coast to sea level rise induced events. (15)
-

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 2nd Term Examination, 2019

URP 4243
(Hazards and Disaster Management)

Full Marks: 210

Time: 3.00 hrs

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume required data.

Section – A

1. (a) Write about the concept "Disaster Risk". Explain the relationship provided in the given equation using appropriate example for each. (10)
"Risk (R) = (H + V + E) - C"
(b) Differentiate between the following aspects of disaster management: (15)
(i) Long term Vs. Short term disaster (ii) Geological Vs. Hydro-metrological disaster (iii) Direct Vs. Indirect losses of disaster
(c) "Risk management and disaster management are two different aspects of management" –justify with required example. (10)
2. (a) Clarify the following aspects of disaster risk management: (05)
(i) Acceptable risk (ii) Residual risk (iii) Risk transfer
(b) "Vulnerability is dynamic" – explain with proper example. Illustrate the double structure framework of vulnerability. Use necessary diagram. (15)
(c) Write short notes on the following issues of disaster management planning: (15)
(i) Contingency planning (ii) Forward planning (iii) Standing Orders on Disaster (SOD)
3. (a) "Urban flooding" is a severe hazard closely associated with unplanned urbanization of Khulna City. Prepare a disaster management cycle suggesting activities to reduce the adverse impact of urban flooding for Khulna City. (10)
(b) Clarify the following aspects of disaster management operation: (15)
(i) Rehabilitation and reconstruction (ii) Resilience and Resistance (iii) Emergency operation plan and contingency
(c) "Evacuation is a vital part of emergency operation" - Write about the problems associated with evacuation procedure. (10)
4. (a) Elaborate the following features of Sendai framework: (15)
(i) Shift from managing disaster to managing risk (ii) People centred approach of disaster management (iii) Multihazard management
(b) Write short notes on the following aspects of Community Based Disaster Risk Management (CBDRM): (05)
(i) Scoping the community (ii) Resource mobilization matrix
(c) Briefly elaborate the process of community risk assessment with appropriate example. (15)

Section – B

5. (a) "Risk is a multidisciplinary spatial problem"- Briefly explain with necessary example. (05)
- (b) Explain with showing the relationship among vulnerability, resilience (capacity to cope) and exposure to hazard to describe vulnerability. (10)
- (c) Describe "Disaster Risk Management Framework" by showing major components to understand risk analysis, risk assessment and risk management. (10)
- (d) Explain the "Risk Governance Framework" in relation to risk assessment and risk management. (10)

6. (a) How does psychology affect in coping with disaster? (05)
- (b) "Disaster related activities of one phase in disaster management cycle have consequences on other phases" - Explain with necessary example (10)
- (c) "Lack of clear national policy may lead the counter disaster management plan to become ineffective" - Explain in short with relevant example. (10)
- (d) Briefly explain the basic issues to be considered while preparing disaster management planning. (10)

7. (a) "Planning and implementation of an existing development plan may be disrupted by disaster and may have impacts on national development" – Briefly Explain. (07)
- (b) "Departments/organizations involved in development have no role in day-to-day disaster management activities" - Explain with necessary example. (08)
- (c) Explain the rationale about having a well-thought link between development and disaster management policy. (10)
- (d) Write down the flood hazards parameters. Briefly explain the approaches of flood damage assessment. (10)

8. (a) Why is wind damage usually higher on the right of the track in the northern hemisphere? (05)
- (b) Explain in short the activities which are covered presently under the Cyclone Preparedness Programme (CPP) taken by Government of Bangladesh and IFRCS. (10)
- (c) Explain GIS applications for hazard assessment with essential example. (10)
- (d) Explain the contexts, applications and issues to be considered while choosing appropriate technology. (10)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 4th Year 2nd Term Examination, 2019

URP 4275

Development Planning

Full Marks: 210

Time: 3.0 hrs

N.B.: i) Answer any **three** questions from each section in separate script

ii) Figures in the right margin indicate marks

iii) Assume data if necessary

Section – A

Marks

1. (a) Differentiate between Development Planning and Development Plan. (08)
(b) Indicative Planning is more realistic and functional than the Central Planning. Evaluate this statement with appropriate global and local level examples. (15)
(c) "Accelerated economic growth and structural change are the two main aims of development planning". Justify it in terms of the pressing challenges and needs in development planning of least developed countries. (12)
2. (a) Briefly discuss the guiding principles of Development Planning in Bangladesh. (10)
(b) Justify that the Articles 9-20 of the State Constitution of Bangladesh have the directives for the formulation of Development Plans in Bangladesh. (10)
(c) Bangladesh has significant achievements in fulfillment of the Vision of the Perspective Plan 2010 -2021. Explain it with the facts and figures. (15)
3. (a) "Development of rural areas with modern urban facilities is very important for the sustainable development of Bangladesh". How will you relate this with the present day slogan My Village - My Town and SDG 11. (10)
(b) Explain the "Productive" and "Resilient" scenarios of Bangladesh Delta Plan. (10)
(c) How will you suggest the improvement of the Economic Lives of the Poor for the sustainable development of Bangladesh as per research findings by the Nobel Laureates Abhijit V. Banerjee and Esther Duflo. (15)
4. Write short notes on the following (any five): (35)
a) Structure of the Planning Commission of Bangladesh b) Institutional Linkages of Planning Commission c) Significance of BDP 2100 for the development of coastal region d) Functions of Physical Infrastructure Division e) Steps in the Formulation of ADP f) Special Economic Zone in Bangladesh g) Approval Process of Investment Projects.

Section – B

5. (a) How the concept of development differs with the concept of economic growth? Explain with necessary example. (08)
(b) Discuss the evolution of the concept of development over time and space. (15)
(c) Describe four important dimensions of modern development. (12)
 6. (a) How the development and globalization are interlinked in the modern world? Discuss with necessary example. (08)
(b) What is development governance? Discuss various types of development governance. (15)
(c) Discuss the WW Rostow's stages of development theory with necessary example. (12)
 7. (a) Illustrate the main features of the core-periphery theory of development. (08)
(b) Compare the balanced growth theory with the unbalanced growth theory with necessary example. (15)
(c) Discuss the social safety net program in Bangladesh and its role in development. (12)
 8. (a) Illustrate the key features of Modernization theory. (08)
(b) Discuss the Todaro's Rural-Urban migration theory with necessary example. (12)
(c) Discuss in brief the industrial policy of Bangladesh. (15)
-